Avocado Price Forecasting for Small Producers: A Comparative Study of Classical, Machine Learning, and Deep Learning Models

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***Abstract*—Accurate price forecasting is critical for agricultural producers to optimize their negotiation strategies. This study presents a comprehensive comparison of ten forecasting models applied to monthly wholesale avocado (Hass) prices in Jalisco, Mexico, over a 6-month horizon. We evaluate classical statistical methods (SARIMA, ETS, SARIMAX), probabilistic ML (Prophet, Gaussian Processes), traditional ML (XGBoost, MLP), and state-of-the-art deep learning architectures (LSTM, N-BEATS, N-HiTS). Our dataset integrates five heterogeneous sources spanning 1996–2025, including government agricultural statistics, wholesale market prices, exchange rates, harvest volumes, and U.S. import demand. Results show that XGBoost with recursive multi-step forecasting achieves the best performance (MAPE 9.03%), significantly outperforming deep learning models (MAPE 48–73%) which suffer from limited training data (97 monthly observations). We find that for short agricultural time series, simpler models with domain-specific feature engineering consistently outperform complex architectures. These findings have practical implications for producer associations like APEAJAL seeking data-driven price negotiation tools.**

***Index Terms*—avocado price forecasting, time series, XGBoost, SARIMA, deep learning, agricultural economics, APEAJAL, Jalisco**

I. Introduction

Mexico produced 2.5 million metric tons of avocado in 2024, accounting for approximately 30% of global production. The country is the world's largest exporter, with the United States as its primary market. Within Mexico, Michoacán dominates production (85.5% of national output), while Jalisco contributes approximately 14.5% with 339,016 tons in 2024 across 28,410 hectares [9].

Despite its smaller share, Jalisco holds a competitive advantage in yield: 11.93 ton/ha compared to Michoacán's 11.16 ton/ha. The Asociación de Productores y Empacadores de Aguacate de Jalisco (APEAJAL) represents these producers, who face significant price volatility in wholesale markets. Over the 2017–2025 period, monthly average prices ranged from \$25.74 to \$95.71 MXN/kg, with a standard deviation of \$16.09 MXN/kg.

This volatility creates an asymmetric information problem: large buyers and intermediaries have access to market intelligence that small producers lack. Accurate price forecasting can help level this playing field by enabling producers to: (1) time their sales to coincide with seasonal price peaks, (2) set minimum acceptable prices during negotiations, and (3) plan harvest schedules and cold storage logistics.

However, agricultural price forecasting in developing countries presents unique challenges. Historical data series are often short (government agencies may have only recently digitized records), data quality varies across sources, and structural market changes—such as trade policy shifts, phytosanitary events, or new export agreements—can invalidate historical patterns.

Recent advances in deep learning have produced models like N-BEATS [2] and N-HiTS [3] that achieve state-of-the-art results on large-scale forecasting benchmarks. However, these models typically require thousands of observations to learn temporal patterns effectively. The question remains: *do these complex architectures outperform simpler methods when applied to short agricultural time series?*

This paper addresses this question through a systematic comparison of 10 forecasting models spanning four paradigms, applied to avocado price prediction in Jalisco. Our contributions are:

- A multi-source integrated dataset combining five heterogeneous data sources for avocado market analysis in Jalisco, spanning 1996–2025.
- A comprehensive benchmark of 10 models from classical statistics to state-of-the-art deep learning, evaluated on 6-month price forecasting.
- Empirical evidence that XGBoost with recursive multi-step forecasting (MAPE 9.03%) significantly outperforms deep learning models (MAPE 48–73%) on short agricultural series.

II. Related Work

A. Agricultural Price Forecasting

Agricultural commodity price forecasting has a long history rooted in ARIMA-family models [4]. Recent work has incorporated machine learning: Xiong et al. applied SVR to agricultural commodity prices, while Ribeiro et al. used ensemble methods for soybean price prediction. For avocado specifically, most studies focus on the U.S. retail market using the Hass Avocado Board dataset [5], analyzing demand elasticity and regional price variations. Mexican wholesale market dynamics, particularly at the state level, remain underexplored in the literature.

B. Deep Learning for Time Series

The landscape of time series forecasting has been transformed by deep learning. LSTM networks [1] introduced gated recurrent architectures capable of learning long-term dependencies. More recently, N-BEATS [2] proposed a pure deep learning architecture based on backward and forward residual links, achieving state-of-the-art results on the M4 competition dataset. N-HiTS [3] extended this with hierarchical interpolation for improved efficiency. The Temporal Fusion Transformer [6] combined attention mechanisms with static and dynamic covariates.

However, Makridakis et al. [7] demonstrated in the M4 competition that statistical methods remain competitive with ML approaches, particularly for shorter series. This finding motivates our investigation of whether the same pattern holds for agricultural price data.

C. Multi-Step Forecasting Strategies

Multi-step forecasting can be approached through direct (separate model per horizon), recursive (iterative one-step predictions), or hybrid strategies [8]. The recursive approach trains a single model to predict one step ahead, then feeds predictions back as inputs for subsequent steps. While this can accumulate errors, it maintains temporal coherence and requires only one model. Ben Taieb et al. found that recursive strategies often outperform direct approaches for shorter horizons.

III. Data and Methodology

A. Data Sources

We integrate five heterogeneous data sources into a unified analytical framework. Table I summarizes each source.

TABLE I
Data sources and characteristics

Source	Variables	Period	Obs.
SNIIM	Wholesale prices	2017–2025	97
SIAP	Production, yield	1996–2024	348
Banxico	USD/MXN rate	2022–2025	1,480
MHAIA	Harvest, shipments	2023–2026	147
HAB/USITC	U.S. demand, prices	2015–2025	570

SNIIM (Sistema Nacional de Información e Integración de Mercados): Monthly wholesale avocado prices across 45 Mexican markets. We compute the Jalisco average as our primary forecasting target.

SIAP (Servicio de Información Agroalimentaria y Pesquera): Annual production statistics disaggregated to monthly estimates using seasonal share patterns. Provides 348 monthly observations from 1996–2024 for Jalisco, including planted area, harvested area, yield, and production volume.

Banxico: Daily USD/MXN exchange rate from the central bank, aggregated to monthly averages. Relevant because avocado export prices are denominated in USD.

MHAIA (Mexican Hass Avocado Importers Association): Weekly harvest and shipment volumes extracted from PDF crop reports. Provides supply-side indicators.

HAB/USITC: U.S. avocado prices and import volumes, representing demand-side signals from Mexico's primary export market.

Fig. 1 shows the complete price series, exhibiting strong seasonality with prices typically peaking in April–June and declining during the harvest season (October–February).

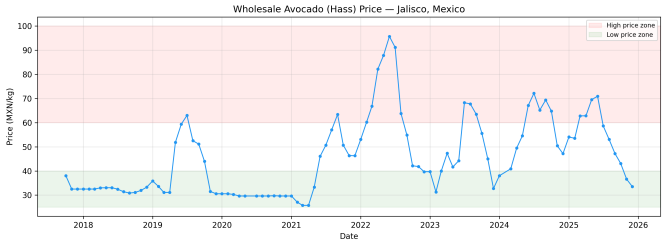

Fig. 1. Monthly wholesale avocado price in Jalisco (2017–2025).

B. Exploratory Analysis

Fig. 2 presents the seasonal price distribution by month. The interquartile range varies significantly: May–July shows the highest median prices ($55–65 MXN/kg), while October–January shows the lowest ($32–40 MXN/kg), coinciding with Jalisco's peak harvest season.

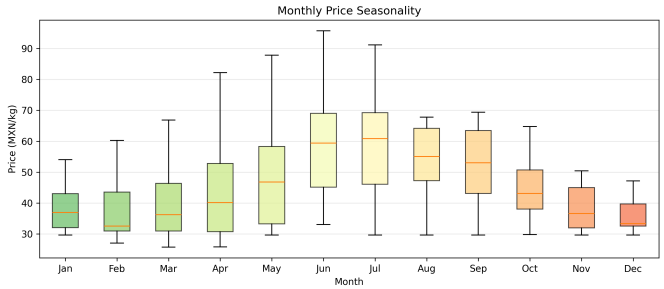

Fig. 2. Seasonal price distribution by month (boxplot).

Key descriptive statistics for the price series: mean $45.96 MXN/kg, standard deviation $16.09, coefficient of variation 35.0%. The series exhibits positive skewness (1.35), indicating occasional price spikes driven by supply shocks.

Fig. 3 shows the long production series from SIAP (1996–2024), revealing a clear growth trend and strong seasonal pattern that SARIMA can exploit.

C. Feature Engineering

For ML models, we construct features from the price series and exogenous variables:

Temporal lags: $y_{t-1}, y_{t-2}, y_{t-3}, y_{t-6}, y_{t-12}$, capturing short and long-term price memory.

Rolling statistics: 3, 6, and 12-month moving averages ($\bar{y}_3, \bar{y}_6, \bar{y}_{12}$) and rolling standard deviation (σ_3), capturing trend and volatility.

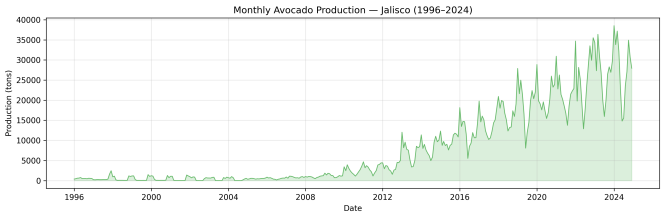

Fig. 3. Monthly avocado production in Jalisco (1996–2024, 348 obs).

Seasonal encoding: Cyclical features $\sin(2\pi m/12)$ and $\cos(2\pi m/12)$ where m is the month, plus quarter indicator. This encoding preserves the circular nature of months (December is close to January).

Exogenous variables: Production volume (tons), U.S. export demand (thousand lbs), average temperature (°C), and import volume (kg), where available.

D. Models Evaluated

We evaluate 10 models across four paradigms:

1) Classical Statistical: **SARIMA** — Seasonal ARIMA with automatic order selection via AIC minimization over $p, q \in \{0, 1, 2\}$ and $P, Q \in \{0, 1\}$, with $d = 1$, $D = 1$, $s = 12$. The model captures both non-seasonal and seasonal differencing:

$$\phi(B)\Phi(B^{12})(1-B)(1-B^{12})y_t = \theta(B)\Theta(B^{12})\epsilon_t \qquad (1)$$

ETS — Exponential Smoothing with automatic selection of error (A/M), trend (A/M/N), and seasonal (A/M) components via AIC. Includes damped trend option.

SARIMAX — SARIMA with exogenous regressors (production, demand, temperature, imports).

2) Probabilistic ML: **Prophet** — Facebook's decomposable model with yearly seasonality, changepoint detection, and configurable prior scales.

Gaussian Processes — Bayesian non-parametric regression with RBF + White noise kernel, providing posterior uncertainty estimates.

3) Traditional ML (Recursive Forecasting): **XGBoost** — Gradient boosted trees (300 estimators, max depth 3, learning rate 0.05). Uses recursive strategy: predict $\hat{y}_{t+1}$, update lag features with $\hat{y}_{t+1}$, predict $\hat{y}_{t+2}$, and iterate for 6 steps.

Ridge — L2-regularized linear regression ($\alpha = 1.0$) with StandardScaler. Serves as the baseline model from the original project, which produced near-flat forecasts.

MLP — Multi-layer perceptron (64-32 hidden units) with StandardScaler preprocessing and early stopping.

4) Deep Learning: **LSTM** — Long Short-Term Memory network (32 hidden units, 12-step input window) via neuralforecast library.

N-BEATS — Neural Basis Expansion Analysis with 12-step input, 500 training steps, standard scaling.

N-HiTS — Neural Hierarchical Interpolation with same configuration as N-BEATS.

E. Evaluation Protocol

We use a temporal train/test split: the last 6 months serve as the test set. We report three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The forecast horizon is 6 months for all models. Statistical models and deep learning use direct multi-step output; ML models use recursive multi-step strategy.

IV. Results

A. Price Forecasting Performance

Table II presents the comparative results for 6-month wholesale price forecasting. Fig. 4 visualizes the MAPE ranking.

TABLE II
COMPARATIVE RESULTS FOR 6-MONTH PRICE FORECASTING

#	Model	MAE	RMSE	MAPE
1	XGBoost	5.24	6.20	9.03%
2	Ridge (Baseline)	8.71	9.53	13.54%
3	Gaussian Process	16.10	18.64	24.24%
3	ETS	14.78	14.93	33.32%
4	Prophet	17.95	18.17	42.15%
5	SARIMA	18.16	18.32	42.48%
6	MLP	26.56	27.70	44.39%
7	LSTM	21.27	21.61	48.42%
8	N-BEATS	24.67	39.19	67.38%
9	N-HiTS	28.09	34.49	73.31%

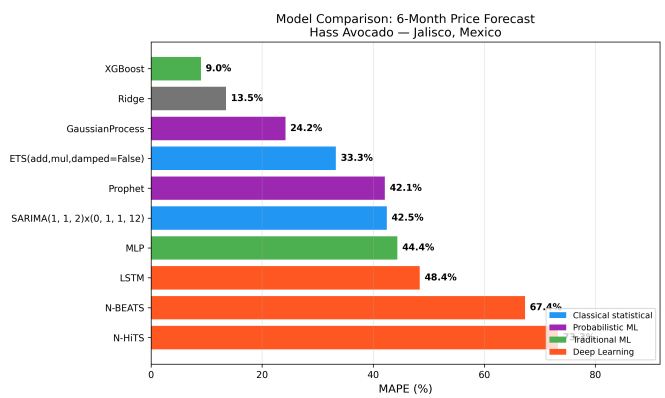

Fig. 4. MAPE comparison across all models, color-coded by paradigm.

XGBoost achieves the best MAPE of 9.03%, nearly 3× better than the third-best model (Gaussian Process, 24.24%). Ridge, the original baseline, achieves a deceptively low MAPE of 13.54%, but its forecast is nearly flat (range of only 3.3 MXN/kg vs. XGBoost's dynamic seasonal range), making it useless for practical decision-making (Fig. 5).

B. Best Model Analysis: XGBoost

Fig. 6 shows XGBoost's train/test fit. The model tracks the test period decline from \$72 to \$47 MXN/kg with reasonable accuracy. Fig. 7 shows the 6-month forecast, which captures

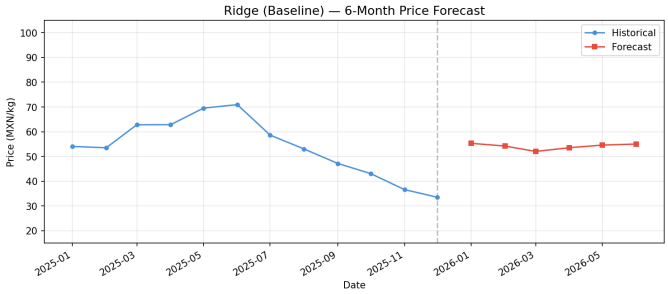

Fig. 5. Ridge (baseline) 6-month forecast: near-flat prediction.

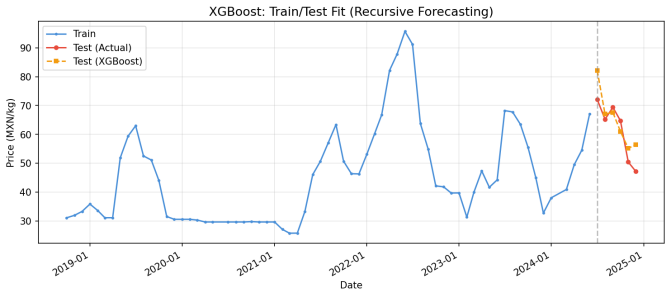

Fig. 6. XGBoost: train/test fit with recursive forecasting.

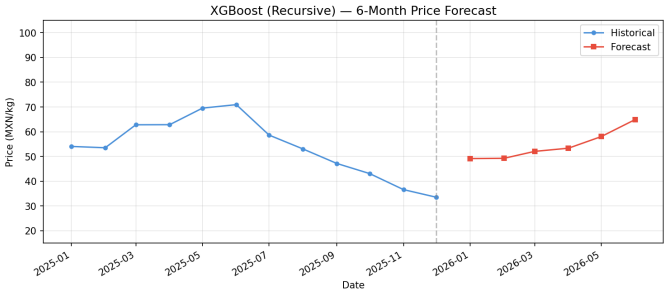

Fig. 7. XGBoost 6-month price forecast showing seasonal recovery.

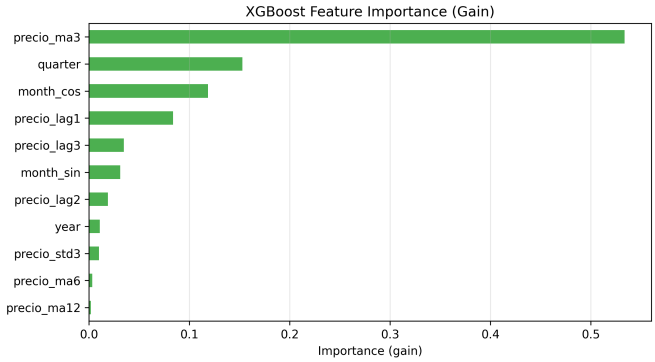

Fig. 8. XGBoost feature importance (gain).

the expected seasonal recovery from \$49 to \$65 MXN/kg—critically, this is *not* a flat line, unlike the Ridge regression baseline that motivated this study.

Fig. 8 shows the feature importance ranking. The most influential features are precio_lag1 (most recent price), precio_ma12 (12-month moving average capturing long-term trend), and the seasonal encodings (month_sin, month_cos). This confirms that the recursive strategy's dynamic lag updates are the key mechanism enabling non-flat multi-step forecasts.

Fig. 9 shows the residual analysis. Residuals are approximately centered at zero with no obvious temporal pattern, though variance increases during the 2022–2023 price spike period, suggesting heteroscedasticity during extreme market conditions.

C. Classical Models

SARIMA and ETS both produce non-flat forecasts that capture seasonal dynamics (Figs. 10–11). ETS with additive trend and multiplicative seasonality performs better (MAPE 33.3%) than SARIMA (42.5%). Both provide 95% confidence intervals, which is valuable for risk-aware decision making—something XGBoost lacks natively.

SARIMAX with exogenous regressors fails catastrophically (MAPE 755%), because the exogenous variables contain interpolated values and incomplete coverage, introducing noise rather than signal.

D. Deep Learning Models

All three deep learning models underperform significantly. LSTM (Fig. 12) shows limited seasonal amplitude in its forecast (\$40–47 range vs expected \$35–65). N-BEATS and N-HiTS (Figs. 13–14) exhibit erratic behavior: initial drops followed by sharp recoveries, inconsistent with historical seasonal patterns.

E. Production Forecasting

For the long production series (348 monthly observations), SARIMA achieves MAPE 9.04% (Fig. 15), confirming that classical models excel when given sufficient historical data. ETS follows at 10.35%, while Prophet underperforms at 20.36%.

TABLE III
PRODUCTION FORECASTING RESULTS (348 OBSERVATIONS)

Model	MAE (tons)	RMSE (tons)	MAPE
SARIMA	2,252	2,994	9.04%
ETS	2,477	2,955	10.35%
Prophet	4,229	5,524	20.36%

V. Discussion

A. Why XGBoost Wins on Short Series

XGBoost's dominance stems from three complementary factors:

Recursive lag updates. By dynamically updating lag features ($y_{t-1}, y_{t-2}, \ldots$) with each prediction, the recursive strategy maintains temporal coherence across the 6-month horizon. This is equivalent to an autoregressive process but with the non-linear modeling capacity of gradient boosted trees.

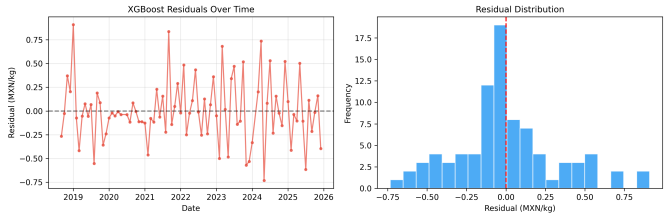

Fig. 9. XGBoost residual analysis: temporal pattern (left) and distribution (right).

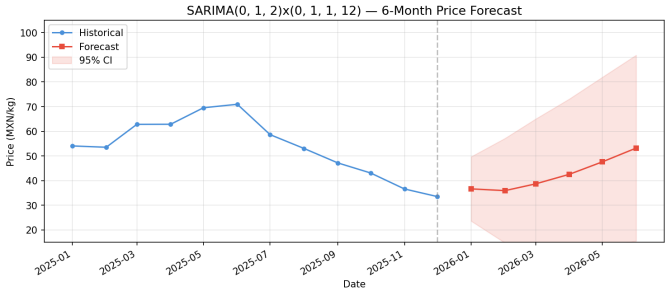

Fig. 10. SARIMA 6-month forecast with 95% confidence interval.

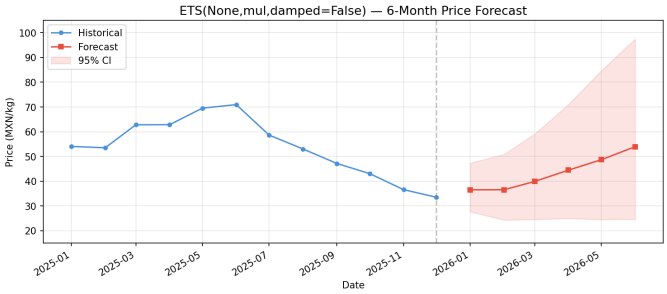

Fig. 11. ETS 6-month forecast with 95% confidence interval.

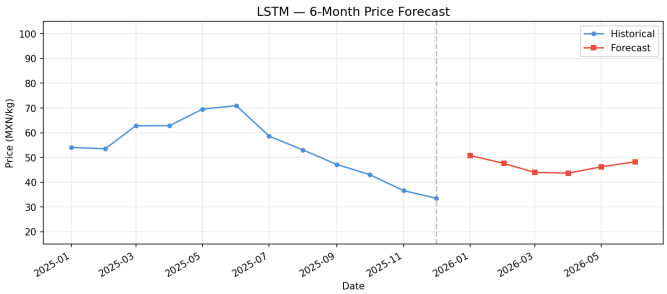

Fig. 12. LSTM 6-month price forecast.

Explicit seasonal encoding. The $\sin/\cos$ month features provide the model with cyclical information "for free," whereas deep learning models must discover this periodicity from raw data—a task that requires many more annual cycles than the 8 available in our series.

Sample efficiency. Tree-based models are inherently sample-efficient: they partition the feature space through binary splits, requiring far fewer observations than neural networks to learn decision boundaries. With 85 training observations (after lag creation), XGBoost has sufficient data to learn meaningful splits, while LSTM/N-BEATS/N-HiTS do not.

B. The Short Series Problem in Deep Learning

Our results align with the M4 competition findings [7]: deep learning models require substantially more data to outperform classical approaches. With 97 monthly observations spanning approximately 8 annual cycles, the deep learning models face a fundamental data scarcity problem:

- **LSTM** (14K parameters): Cannot learn robust seasonal representations from 8 cycles. The model defaults to conservative, low-amplitude predictions.
- **N-BEATS** (2.4M parameters): Severely overparameterized for 97 observations. The model memorizes training patterns but fails to generalize.
- **N-HiTS**: Similar overparameterization issues, with erratic multi-step behavior.

Interestingly, when SARIMA is given the long production series (348 observations, 28 annual cycles), it achieves excellent performance (MAPE 9.04%), confirming that the issue is data quantity, not model capability.

C. The SARIMAX Failure

SARIMAX's catastrophic failure (MAPE 755%) deserves attention. The exogenous variables—production, demand, temperature, imports—have incomplete temporal coverage (28–98% of months). Interpolation and forward-filling introduce artificial patterns that the model treats as signal. This highlights a critical lesson: *multivariate models are only as good as their weakest data source*. For practitioners, it is better to use a well-specified univariate model than a multivariate model with poor-quality covariates.

D. Practical Implications for APEAJAL

Based on our results, we recommend a two-model approach for APEAJAL:

1) **XGBoost** for point forecasts: Best accuracy (MAPE 9.03%), captures seasonal dynamics, suitable for setting target prices in negotiations.
2) **ETS** for uncertainty quantification: Provides 95% confidence intervals that help producers understand forecast risk and set minimum acceptable prices.

The seasonal pattern identified—prices peaking in May–July and declining October–February—suggests that Jalisco producers should prioritize sales during the off-season months when supply is lower and prices are higher, potentially using cold storage to shift sales timing.

E. Limitations and Future Work

Several limitations should be noted: (1) the test period (Jul–Dec 2025) coincided with an atypical price decline, potentially inflating MAPE for models predicting seasonal recovery; (2) no time series cross-validation was performed due to the

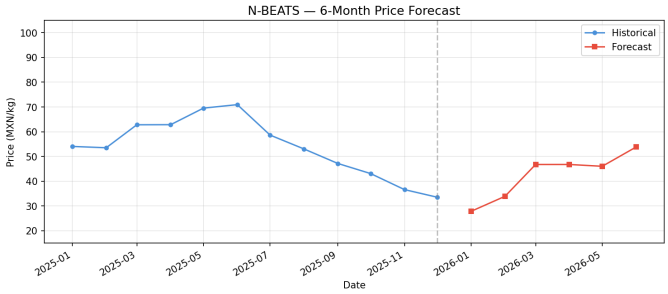

Fig. 13. N-BEATS 6-month price forecast.

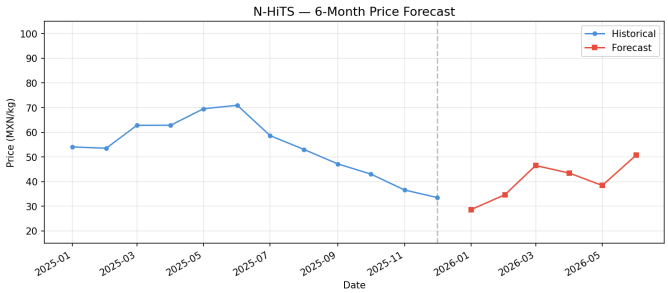

Fig. 14. N-HiTS 6-month price forecast.

short series; (3) the SIAP price data proved to be annual rather than truly monthly; (4) deep learning models were trained with default hyperparameters—extensive tuning might improve their performance.

Future work should explore: ensemble methods combining XGBoost accuracy with ETS uncertainty quantification; transfer learning from longer related series (e.g., Michoacán prices); and the incorporation of real-time satellite imagery for crop condition monitoring as an exogenous variable.

VI. Conclusions

This study demonstrates that for short agricultural time series, traditional machine learning with careful feature engineering outperforms both classical statistical methods and state-of-the-art deep learning architectures. XGBoost with recursive multi-step forecasting achieves MAPE 9.03% on 6-month avocado price prediction in Jalisco, compared to 33–42% for classical models and 48–73% for deep learning.

The key insight is that *model complexity does not compensate for data scarcity*. With only 97 monthly observations, explicit feature engineering (temporal lags, rolling statistics, seasonal encoding) provides more value than the implicit pattern learning of deep neural networks. This finding has broad implications for agricultural forecasting in developing countries, where short data series are the norm rather than the exception.

For APEAJAL and similar producer associations, our results provide a practical, implementable forecasting tool that can support data-driven price negotiations and sales timing decisions.

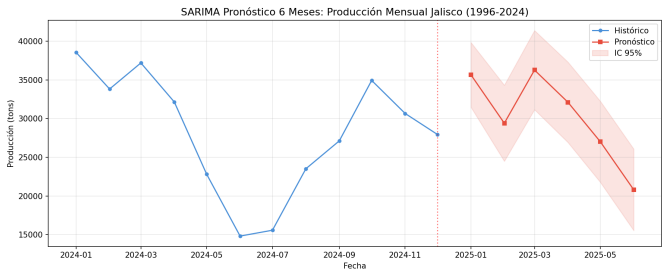

Fig. 15. SARIMA production forecast (348-month series). MAPE 9.04%.

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